Project Specification

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Itinera - AI-Powered Smart Travel Itinerary Management Platform

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Introduction

Travel planning is often fragmented across emails, booking confirmations, calendars, screenshots, and notes. Users frequently need to manually organize flight confirmations, hotel reservations, restaurant bookings, and activities into a single coherent itinerary. This process is repetitive, time-consuming, and prone to errors.

Itinera is an AI-powered travel planning platform designed to simplify itinerary management by automatically converting forwarded booking emails into structured itinerary items. Users can visually manage trips through an interactive timeline interface while also interacting with the system conversationally using natural language commands.

The platform combines artificial intelligence, structured data extraction, conversational workflow management, and dynamic itinerary rendering to create a streamlined travel planning experience.

The primary benefit of the system is reducing manual trip organization work while improving accessibility and convenience for travelers.

Objectives

The objectives of this project are:

1. Develop a centralized travel itinerary management PWA
2. Allow users to manage multiple trips
3. Implement AI-powered email parsing for automatic itinerary generation.

4. Build a conversational AI assistant capable of modifying itineraries using natural language.
5. Create a dynamic visual timeline interface for trip organization.
6. Demonstrate practical applications of AI tool-calling architectures and structured information extraction.
7. Provide an intuitive and user-friendly workflow for travel planning.

Solution Methodology

Software / Hardware Requirements

Software

- Frontend: SvelteKit
- Styling: Tailwind CSS + shadcn-svelte
- Backend: Node.js + Express + Typescript
- Database: Supabase PostgreSQL
- Authentication: Supabase Auth
- AI Integration: Venice AI
- Email Handling: Resend or SendGrid Inbound Parse

System Implementation Plan

1. MVP User Authentication System

Users will create accounts and securely log in to manage personal trips. Authentication sessions will persist across visits.

Features:

- User sign up
- Login/logout
- Secure session management

2. Trip Management Module

Users will be able to:

- Create trips
- Edit trip details
- Delete trips
- View trip information

Trip information includes:

- Trip name
- Destination
- Start and end dates
- Description

Trips have 1..n activities.

3. Activity CRUD System

The system will support itinerary item management including:

- Create itinerary items
- Edit itinerary items
- Delete itinerary items
- Reorder itinerary items

Supported itinerary types:

- Flights
- Hotels
- Restaurants
- Activities/Events
- Transportation

Each activity has exactly 1 associated trip.

4. Timeline Interface

A visual itinerary timeline will display all activities chronologically.

Features:

- Day grouping
- Chronological sorting
- Expandable activity cards
- Interactive trip visualization

This component serves as one of the primary user experience features of the application.

5. AI Conversational Assistant

Users can modify activities using natural language requests such as:

- “Move dinner after the concert”
- “Add breakfast at 9am”
- “Reschedule museum visit”

The AI assistant converts user instructions into backend CRUD operations using tool-calling workflows.

6. Tool Calling Architecture

Backend functions will support AI-generated actions.

Example tools:

- createTrip()
- createActivity()
- updateActivity()
- deleteActivity()
- reorderActivities()

This architecture demonstrates practical AI workflow automation.

7. Email Forwarding and Ingestion

Users will forward booking emails to a designated application email address.

Example:
plans@itinera.app

The backend system will:

1. Receive email contents
2. Extract relevant text
3. Send the text to an LLM parser
4. Generate structured itinerary data

8. AI Information Extraction

The AI model will extract structured travel information from:

- Flight confirmations
- Hotel bookings
- Reservation emails
- Event confirmations

Example structured output:

```
{
 "type": "flight",
 "title": "Air Canada Flight",
 "departure": "Toronto",
 "arrival": "Vancouver",
 "departureTime": "2026-07-14T08:30"
}
```

9. Automatic Activity Generation and Trip Mapping

After extracting structured information:

- The system identifies the correct trip
- Creates activity entries automatically
- Updates the visual timeline

This is the primary “smart automation” feature of the platform.

Validation Strategy

The system will be validated through:

- User workflow testing
- Functional testing:
    - CRUD operations
    - AI extraction accuracy testing
    - Timeline rendering
    - Authentication and sessions
    - Conversational AI interaction

Performance will be measured based on:

- Extraction accuracy
- Activity generation success rate
- User interaction responsiveness

Ethics and Sustainability Considerations

Ethical Design

The platform will:

- Protect user travel and personal information by leveraging encryption
- Use secure authentication methods
- Avoid storing unnecessary sensitive data

Human-Centered Design

The interface will prioritize:

- Simplicity
- Accessibility
- Clear user workflows
- Reduced manual workload

Equity, Diversity, and Inclusivity

The platform aims to provide:

- Simple and understandable interfaces
- Accessible interaction patterns
- Support for diverse user travel needs

Sustainability

Cloud infrastructure and managed backend services help reduce unnecessary local hardware usage while enabling scalable deployment.

Commercialization Considerations

1. Importance to Users (8/10)

Travel organization is a common problem for many users, especially frequent travelers who manage multiple bookings across different platforms.

2. Existing Competition (4/10)

There are existing itinerary applications; however, relatively few provide AI-powered automatic email extraction combined with conversational itinerary editing.

3. Why Users Would Choose Itinera

- **Easier to Use:** Automatically converts booking emails into organized activity entries.
- **Comprehensive Solution:**
    - AI extraction
    - Timeline visualization
    - Conversational editing into a unified workflow.

Other Advantages: Reduces manual trip planning effort and improves organization efficiency.

Proposed Timeline

No.	Activity	Week #
1	Requirements gathering and project planning	1
2	Database schema and backend setup	1–2
3	Authentication implementation	2
4	Trip and Activity CRUD Tool Development	2–3
5	Timeline UI development	4
6	AI assistant chat interface and text streaming	4–5
7	Email ingestion and parsing system	5
8	LLM structured extraction implementation	5–6
9	Automatic itinerary generation workflows	6
10	Testing and debugging	7
11	Final polishing and presentation preparation	8